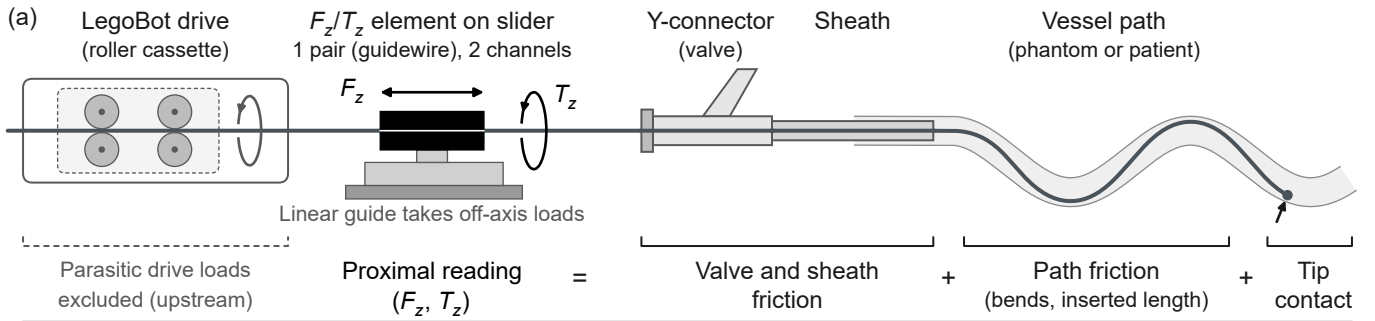


Proximal reading − per-run friction baseline (valve-only and per-segment traces) = F_z excess → LegoBot limits (set from bench baselines)

